# Evaluation runs: lock (a3-lock)

generated 2026-11-03 from brain-data 03b0b57cc 2026-11-03 11:21:51 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): fifo: expected conforms; control-lifo: expected conforms; wrong-round: expected violation; double-grant: expected violation; double-grant: expected conforms (one client); never-grant: expected violation; never-grant: expected conforms (one client). Blue rows are at the current pins (TraceForge 9a0629e, merge join 367cdc6, JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G^{expl}_{spec}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕒 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair control-lifo, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 28 | conforms | 0.0 | 2 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-02 |
| 2,3 | 20 | 212 | conforms | 0.1 | 4 | 1 h, 250 GB | 20 | 210 | brain03, 2026-11-02 |
| 2,4 | 70 | 1676 | conforms | 0.2 | 2 | 1 h, 250 GB | 75 | 2625 | brain03, 2026-11-02 |
| 2,5 | 252 | 13604 | conforms | 0.9 | 2 | 1 h, 250 GB | 516 | 65867 | brain03, 2026-11-02 |
| 2,6 | 924 | 112380 | conforms | 9.4 | 4 | 1 h, 250 GB | 2424 | 1192017 | brain03, 2026-11-02 |
| 3,2 | 90 | 2850 | conforms | 0.3 | 2 | 1 h, 250 GB | 98 | 4366 | brain03, 2026-11-02 |
| 3,3 | 1680 | 324282 | conforms | 26.7 | 23 | 1 h, 250 GB | 3214 | 2658596 | brain03, 2026-11-02 |
| 3,4 | 34650 | | 🕒 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,2 | 2520 | 651168 | conforms | 60.6 | 22 | 1 h, 250 GB | 4833 | 5900122 | brain03, 2026-11-02 |
| 4,3 | | | 🕒 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 5,2 | | | 🕒 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair fifo, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 28 | conforms | 0.0 | 2 | 1 h, 250 GB | 6 | 21 | brain03, 2026-11-02 |
| 2,3 | 20 | 212 | conforms | 0.0 | 2 | 1 h, 250 GB | 20 | 210 | brain03, 2026-11-02 |
| 2,4 | 70 | 1676 | conforms | 0.2 | 2 | 1 h, 250 GB | 77 | 2656 | brain03, 2026-11-02 |
| 2,5 | 252 | 13604 | conforms | 0.9 | 2 | 1 h, 250 GB | 520 | 66350 | brain03, 2026-11-02 |
| 2,6 | 924 | 112380 | conforms | 9.5 | 2 | 1 h, 250 GB | 2505 | 1201363 | brain03, 2026-11-02 |
| 3,2 | 90 | 2898 | conforms | 0.3 | 2 | 1 h, 250 GB | 129 | 5402 | brain03, 2026-11-02 |
| 3,3 | 1680 | 330252 | conforms | 21.2 | 24 | 1 h, 250 GB | 4560 | 3790449 | brain03, 2026-11-02 |
| 3,4 | 34650 | | 🕒 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 4,2 | 2520 | 675960 | conforms | 60.2 | 22 | 1 h, 250 GB | 8437 | 10077551 | brain03, 2026-11-02 |
| 4,3 | | | 🕒 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |
| 5,2 | | | 🕒 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair double-grant, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 70 | 64 | violation | 0.0 | 4 | 1 h, 250 GB | 70 | 405 | brain03, 2026-11-02 |
| 3,2 | 90 | 34650 | 34560 | violation | 31.8 | 38 | 1 h, 250 GB | 34650 | 3114495 | brain03, 2026-11-02 |
| 3,3 | 1680 | | | 🕒 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair **never-grant**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 21 | 8 | violation | 0.0 | 4 | 1 h, 250 GB | 14 | 69 | brain03, 2026-11-02 |
| 3,2 | 90 | 1504 | 716 | violation | 2.3 | 4 | 1 h, 250 GB | 806 | 68535 | brain03, 2026-11-02 |
| 3,3 | 1680 | 83852 | 40576 | violation | 346.1 | 36 | 1 h, 250 GB | 42256 | 69579720 | brain03, 2026-11-02 |

### stateless: pair **wrong-round**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{spec}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 28 | 28 | violation | 0.0 | 4 | 1 h, 250 GB | 28 | 168 | brain03, 2026-11-02 |
| 3,2 | 90 | 2898 | 2898 | violation | 6.2 | 4 | 1 h, 250 GB | 2898 | 260820 | brain03, 2026-11-02 |
| 3,3 | 1680 | 330252 | 330252 | violation | 2820.5 | 98 | 1 h, 250 GB | 330252 | 554823360 | brain03, 2026-11-02 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): double-grant 2 3 fifo, never-grant 2 3 fifo, wrong-round 2 3 fifo, double-grant 2 4 fifo, wrong-round 2 4 fifo, never-grant 2 4 fifo, double-grant 2 5 fifo, wrong-round 2 5 fifo, never-grant 2 5 fifo, never-grant 2 6 fifo, double-grant 2 6 fifo, wrong-round 2 6 fifo, wrong-round 3 4 fifo, never-grant 3 4 fifo, never-grant 3 5 fifo, wrong-round 3 5 fifo, control-lifo 3 5 fifo, fifo 3 5 fifo, never-grant 3 6 fifo, wrong-round 3 6 fifo, fifo 3 6 fifo, control-lifo 3 6 fifo, wrong-round 4 2 fifo, never-grant 4 2 fifo, … (91 instances in all).

skipped: a smaller size had reached its limit: double-grant 3 4 fifo, double-grant 3 5 fifo, double-grant 3 6 fifo.

## stateful

### stateful: pair **control-lifo**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 28 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 20 | 212 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 70 | 1676 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5 | 252 | 13604 | conforms | 0.3 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,6 | 924 | 112380 | conforms | 1.3 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2 | 90 | 2850 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 1680 | 324282 | conforms | 2.6 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 34650 | 40400514 | conforms | 782.0 | 188 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 2520 | 651168 | conforms | 6.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

### stateful: pair **fifo**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 28 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 20 | 212 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 70 | 1676 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5 | 252 | 13604 | conforms | 0.3 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,6 | 924 | 112380 | conforms | 1.2 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2 | 90 | 2898 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 1680 | 330252 | conforms | 2.8 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 34650 | 41133774 | conforms | 779.4 | 196 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 2520 | 675960 | conforms | 6.1 | 6 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

### stateful: pair **double-grant**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 70 | 64 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 20 | 924 | 904 | violation | 0.1 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 70 | 12870 | 12800 | violation | 0.5 | 28 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5 | 252 | 184756 | 184504 | violation | 3.2 | 294 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,6 | 924 | 2704156 | 2703232 | violation | 42.5 | 4282 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2 | 90 | 34650 | 34560 | violation | 0.6 | 48 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 1680 | 17153136 | 17151456 | violation | 171.9 | 15095 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 2520 | 63063000 | 63060480 | violation | 881.9 | 41278 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

### stateful: pair **never-grant**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 21 | 8 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 20 | 106 | 43 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 70 | 545 | 224 | violation | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5 | 252 | 2864 | 1181 | violation | 0.2 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,6 | 924 | 15309 | 6320 | violation | 0.5 | 12 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2 | 90 | 1504 | 716 | violation | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 1680 | 83852 | 40576 | violation | 1.0 | 34 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 34650 | 5077092 | 2448528 | violation | 83.2 | 1062 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 2520 | 257844 | 136092 | violation | 2.9 | 164 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2 | 113400 | 83803176 | 47273640 | violation | 3204.8 | 54125 | 1 h, 250 GB | brain02, 2026-11-02 |

### stateful: pair **wrong-round**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 28 | 28 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,3 | 20 | 212 | 212 | violation | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,4 | 70 | 1676 | 1676 | violation | 0.2 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,5 | 252 | 13604 | 13604 | violation | 0.4 | 6 | 1 h, 250 GB | brain02, 2026-11-02 |
| 2,6 | 924 | 112380 | 112380 | violation | 1.7 | 14 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,2 | 90 | 2898 | 2898 | violation | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,3 | 1680 | 330252 | 330252 | violation | 2.9 | 116 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,4 | 34650 | 41133774 | 41133774 | violation | 556.5 | 6437 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3,5 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,2 | 2520 | 675960 | 675960 | violation | 7.2 | 602 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4,3 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |
| 5,2 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): never-grant 3 6 fifo, wrong-round 3 6 fifo, double-grant 4 4 fifo, wrong-round 4 4 fifo, never-grant 4 4 fifo, wrong-round 4 5 fifo, never-grant 4 5 fifo, wrong-round 4 6 fifo, never-grant 4 6 fifo, wrong-round 5 3 fifo, never-grant 5 3 fifo, double-grant 5 3 fifo, never-grant 5 4 fifo, wrong-round 5 4 fifo, double-grant 5 4 fifo, wrong-round 5 5 fifo, never-grant 5 5 fifo, double-grant 5 5 fifo, wrong-round 5 6 fifo, never-grant 5 6 fifo, double-grant 5 6 fifo, double-grant 6 2 fifo, never-grant 6 2 fifo, wrong-round 6 2 fifo, … (38 instances in all).

skipped: a smaller size had reached its limit: double-grant 3 5 fifo, fifo 3 6 fifo, control-lifo 3 6 fifo, double-grant 3 6 fifo, fifo 4 4 fifo, double-grant 4 4 fifo, control-lifo 4 4 fifo, control-lifo 4 5 fifo, fifo 4 5 fifo, double-grant 4 5 fifo, control-lifo 4 6 fifo, fifo 4 6 fifo, double-grant 4 6 fifo, fifo 5 3 fifo, control-lifo 5 3 fifo, control-lifo 5 4 fifo, fifo 5 4 fifo, fifo 5 5 fifo, control-lifo 5 5 fifo, control-lifo 5 6 fifo, fifo 5 6 fifo, fifo 6 2 fifo, control-lifo 6 2 fifo, control-lifo 6 3 fifo, … (31 instances in all).

## offline

### offline: pair **control-lifo**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 28 | conforms | 1.0 | 4 | 10 min, 16 GB | 0.6 | 0.4 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,2 | 90 | 2850 | conforms | 0.5 | 4 | 10 min, 16 GB | 0.2 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 1680 | 324282 | conforms | 5.0 | 32 | 1 h, 250 GB | 0.5 | 3.8 | 0.7 | merge | brain01, 2026-11-03 |

### offline: pair **fifo**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 28 | conforms | 0.7 | 4 | 10 min, 16 GB | 0.6 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,2 | 90 | 2898 | conforms | 0.5 | 4 | 10 min, 16 GB | 0.2 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 1680 | 330252 | conforms | 4.8 | 32 | 1 h, 250 GB | 0.5 | 3.5 | 0.7 | merge | brain01, 2026-11-03 |

### offline: pair **double-grant**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 70 | 28 | violation | 0.7 | 4 | 10 min, 16 GB | 0.6 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,2 | 90 | 34650 | 7044 | violation | 1.3 | 12 | 1 h, 250 GB | 0.2 | 0.9 | 0.2 | merge | brain01, 2026-11-03 |
| 3,3 | 1680 | 17153136 | 1823232 | violation | 287.9 | 4650 | 1 h, 250 GB | 0.5 | 186.1 | 101.3 | merge | brain01, 2026-11-03 |

### offline: pair **never-grant**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 21 | 5 | violation | 0.7 | 4 | 10 min, 16 GB | 0.6 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,2 | 90 | 1504 | 176 | violation | 0.4 | 4 | 10 min, 16 GB | 0.2 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 1680 | 83852 | 4076 | violation | 2.1 | 76 | 10 min, 16 GB | 0.5 | 1.2 | 0.4 | in-memory | brain01, 2026-11-02 |

### offline: pair **wrong-round**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 6 | 28 | 6 | violation | 0.7 | 4 | 10 min, 16 GB | 0.6 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,2 | 90 | 2898 | 372 | violation | 0.6 | 4 | 10 min, 16 GB | 0.2 | 0.4 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3,3 | 1680 | 330252 | 13584 | violation | 5.0 | 147 | 10 min, 16 GB | 0.5 | 3.8 | 0.7 | in-memory | brain01, 2026-11-02 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): double-grant 2 3 fifo, never-grant 2 3 fifo, control-lifo 2 3 fifo, fifo 2 3 fifo, wrong-round 2 3 fifo, double-grant 2 4 fifo, fifo 2 4 fifo, control-lifo 2 4 fifo, wrong-round 2 4 fifo, never-grant 2 4 fifo, double-grant 2 5 fifo, wrong-round 2 5 fifo, never-grant 2 5 fifo, control-lifo 2 5 fifo, fifo 2 5 fifo, never-grant 2 6 fifo, control-lifo 2 6 fifo, fifo 2 6 fifo, double-grant 2 6 fifo, wrong-round 2 6 fifo, wrong-round 3 4 fifo, double-grant 3 4 fifo, never-grant 3 4 fifo, control-lifo 3 4 fifo, … (110 instances in all).

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **control-lifo**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 28 | 28 | conforms | 4.5 | 168 | 10 min, 16 GB | 4.1 | 0.4 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,2 | | 2850 | conforms | 0.0 | 4 | 10 min, 250 GB | | 0.3 | 0.0 | merge | brain02, 2026-11-03 |
| 3,3 | | 324282 | conforms | 1.1 | 32 | 10 min, 250 GB | | 3.8 | 1.1 | merge | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **double-grant**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | | 70 | conforms | 0.0 | 4 | 10 min, 250 GB | | 0.2 | 0.0 | merge | brain02, 2026-11-03 |
| 3,2 | | 34650 | conforms | 0.5 | 8 | 10 min, 250 GB | | 0.9 | 0.5 | merge | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **fifo**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 28 | 28 | conforms | 4.4 | 150 | 10 min, 16 GB | 4.3 | 0.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,2 | | 2898 | conforms | 0.2 | 4 | 10 min, 250 GB | | 0.3 | 0.2 | merge | brain02, 2026-11-03 |
| 3,3 | | 330252 | conforms | 1.4 | 32 | 10 min, 250 GB | | 3.5 | 1.4 | merge | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **never-grant**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 21 | 21 | conforms | 5.2 | 154 | 10 min, 16 GB | 5.1 | 0.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,2 | | 1504 | conforms | 0.0 | 4 | 10 min, 250 GB | | 0.2 | 0.0 | merge | brain02, 2026-11-03 |
| 3,3 | | 83852 | conforms | 1.0 | 14 | 10 min, 250 GB | | 1.2 | 1.0 | merge | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **wrong-round**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 28 | 28 | conforms | 4.4 | 164 | 10 min, 16 GB | 4.2 | 0.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,2 | | 2898 | conforms | 0.0 | 4 | 10 min, 250 GB | | 0.4 | 0.0 | merge | brain02, 2026-11-03 |
| 3,3 | | 330252 | conforms | 1.2 | 32 | 10 min, 250 GB | | 3.8 | 1.2 | merge | brain02, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** refused (an input missing at run time): double-grant 3 3 fifo.jimpl-tfimpl.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **control-lifo**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 28 | 28 | conforms | 4.5 | 168 | 10 min, 16 GB | 0.4 | 4.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,2 | 2850 | | conforms | 0.0 | 4 | 10 min, 250 GB | 0.3 | | 0.0 | merge | brain02, 2026-11-03 |
| 3,3 | 324282 | | conforms | 1.4 | 32 | 10 min, 250 GB | 3.8 | | 1.4 | merge | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **double-grant**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 70 | | conforms | 0.0 | 4 | 10 min, 250 GB | 0.2 | | 0.0 | merge | brain02, 2026-11-03 |
| 3,2 | 34650 | | conforms | 0.5 | 8 | 10 min, 250 GB | 0.9 | | 0.5 | merge | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **fifo**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 28 | 28 | conforms | 4.4 | 150 | 10 min, 16 GB | 0.1 | 4.3 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,2 | 2898 | | conforms | 0.0 | 4 | 10 min, 250 GB | 0.3 | | 0.0 | merge | brain02, 2026-11-03 |
| 3,3 | 330252 | | conforms | 1.3 | 32 | 10 min, 250 GB | 3.5 | | 1.3 | merge | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **never-grant**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 21 | 21 | conforms | 5.2 | 154 | 10 min, 16 GB | 0.2 | 5.1 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,2 | 1504 | | conforms | 0.0 | 4 | 10 min, 250 GB | 0.2 | | 0.0 | merge | brain02, 2026-11-03 |
| 3,3 | 83852 | | conforms | 0.6 | 14 | 10 min, 250 GB | 1.2 | | 0.6 | merge | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **wrong-round**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,2 | 28 | 28 | conforms | 4.4 | 164 | 10 min, 16 GB | 0.2 | 4.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3,2 | 2898 | | conforms | 0.0 | 4 | 10 min, 250 GB | 0.4 | | 0.0 | merge | brain02, 2026-11-03 |
| 3,3 | 330252 | | conforms | 1.2 | 32 | 10 min, 250 GB | 3.8 | | 1.2 | merge | brain02, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** skipped: a smaller size had reached its limit: double-grant 3 3 fifo.tfimpl-jimpl.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **control-lifo**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 28 | 28 | conforms | 3.6 | 164 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,2 | 2850 | 2850 | conforms | 10.3 | 1292 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 324282 | 324282 | conforms | 387.5 | 14271 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **double-grant**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 70 | 70 | conforms | 5.6 | 175 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,2 | 34650 | 34650 | conforms | 61.5 | 7169 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 17153136 | | (1 h) | | | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **fifo**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 28 | 28 | conforms | 3.6 | 155 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,2 | 2898 | 2898 | conforms | 12.3 | 2959 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 330252 | 330252 | conforms | 485.6 | 13798 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **never-grant**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 21 | 21 | conforms | 3.6 | 158 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,2 | 1504 | 1504 | conforms | 10.7 | 2646 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 83852 | 83852 | conforms | 146.7 | 7630 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **wrong-round**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 28 | 28 | conforms | 3.5 | 166 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,2 | 2898 | 2898 | conforms | 10.5 | 3184 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 330252 | 330252 | conforms | 395.1 | 13563 | 1 h, 250 GB | | brain02, 2026-11-03 |

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **control-lifo**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 28 | 28 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,2 | 2850 | 2850 | conforms | 0.2 | 6 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 324282 | 324282 | conforms | 4.7 | 136 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **double-grant**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 70 | 70 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,2 | 34650 | 34650 | conforms | 1.1 | 48 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **fifo**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 28 | 28 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,2 | 2898 | 2898 | conforms | 0.2 | 6 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 330252 | 330252 | conforms | 4.9 | 143 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **never-grant**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 21 | 21 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,2 | 1504 | 1504 | conforms | 0.1 | 6 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 83852 | 83852 | conforms | 1.6 | 70 | 1 h, 250 GB | | brain02, 2026-11-03 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair wrong-round, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2,2 | 28 | 28 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,2 | 2898 | 2898 | conforms | 0.2 | 6 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3,3 | 330252 | 330252 | conforms | 6.7 | 139 | 1 h, 250 GB | | brain02, 2026-11-03 |

**Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).**
skipped: a smaller size had reached its limit: double-grant 3 3 fifo.